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Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

1. TITLE: Velocity-Responsive Bioluminescent Plant Matrix for Adaptive Urban Illumination Systems.

2. INTERNAL INVENTOR(S)/STUDENT(S): All fields in this column are mandatory to be filled

3. DESCRIPTION OF THE INVENTION:

The present invention is directed to the field of street/pathway lighting and, more specifically, to a living lighting matrix of light-emitting plant modules whose intensity is varied based on the speed of detected approaching pedestrians, cyclists, and/or vehicles to conserve energy while still providing adequate illumination. Said invention combines advances in synthetic biology (engineered plants) with embedded motion/velocity sensing and control to create a low-power alternative to conventional electric street lights/lights.

Conventional street/pathway lighting is predominantly electric and saps energy from the grid, with little to no power-saving capacity when no pedestrian or vehicle is detected due to constant operation at a set intensity. Recent advances in synthetic biology and specifically the availability of commercial autofluorescent ornamental plants that utilize fungal bioluminescence suggest that engineered plant luminescence could provide a eco-friendly option for street lighting; however, such glow-in-the-dark plants provide only a very low level of light intensity that is constant regardless of the needs of the environment/lighting demands. (Kanjapokin et al., 2023)

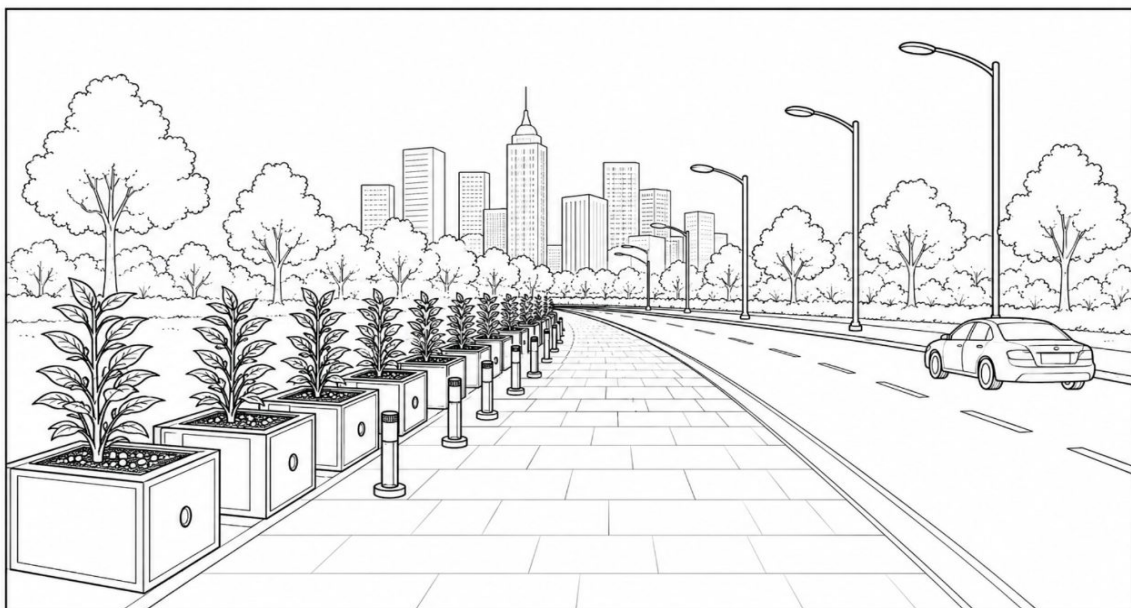


Figure 1. Overall architecture of the proposed system.

A. PROBLEM ADDRESSED BY THE INVENTION:

Conventional electric urban lighting consumes continuous grid or stored energy and contributes to carbon emissions and light pollution even in low-traffic periods.

Existing "smart" motion-responsive street lighting systems successfully reduce energy use by reacting to presence or velocity of road users, but remain entirely dependent on electrically powered luminaires, wiring infrastructure, and grid connectivity. (Connected Streetlighting Systems, 2024)

Existing bioluminescent plant technologies produce a fixed, ambient, non-adjustable glow intensity that is not bright enough for practical illumination and includes no means of external, real-time modulation tied to approach velocity. (Group, 2023)

There is presently no known system combining a living bioluminescent organism with an embedded velocity-sensing and modulation layer to produce urban illumination that brightens proportionally to the speed and nearness of an approaching road user, and dims automatically when the area is unoccupied. (Decardi-Nelson & You, 2024)

B. OBJECTIVE OF THE INVENTION

To provide a bioluminescent plant-based illumination matrix in which individual plant modules can have their luminescence intensity actively boosted above baseline constitutive glow through a coupled bio-stimulation or hybrid opto-biological mechanism.

To provide a velocity-sensing subsystem (radar, LIDAR, or passive-infrared/Doppler based) capable of detecting the presence, direction, and approach speed of a pedestrian, cyclist, or vehicle.

To provide a control layer that maps detected velocity to a corresponding luminescence-boost signal, with faster-approaching users triggering a wider, more anticipatory response and slower pedestrians triggering a localized, lower-intensity response.

To reduce net grid electricity consumption for low-and-moderate intensity urban lighting by supplementing electric luminaires with a self-sustaining biological light source.

To provide a modular, replantable matrix architecture such that individual plant units can be replaced or maintained independently without disrupting the sensing/control infrastructure.

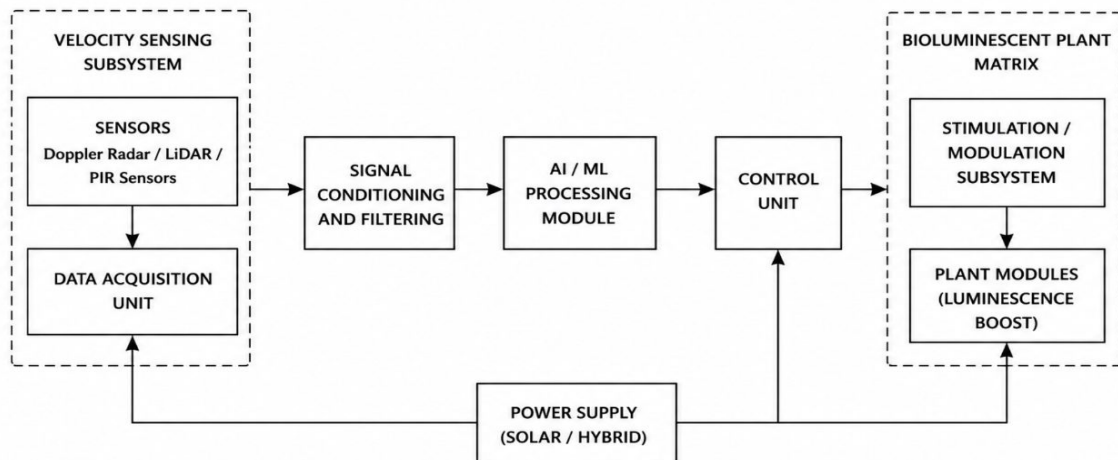


Figure 2. Block diagram of sensing and illumination control.

C. STATE OF THE ART/RESEARCH GAP/NOVELTY: Describe how your invention fulfills the research gap?

FEATURE	CONVENTIONAL ELECTRIC LIGHTING	SMART ELECTRIC LIGHTING (MOTION/VELOCITY BASED)	EXISTING BIOLUMINESCENT PLANTS	PROPOSED INVENTION
ENERGY SOURCE	Grid Electricity	Grid / Solar Electricity	Metabolic Energy (Plant)	Metabolic Energy (Plant) + Minimal Auxiliary Power
RESPONSIVENESS	None	Motion / Velocity Based	None (Constant Glow)	Velocity-Proportional Dynamic Response
BRIGHTNESS CONTROL	Fixed	Adjustable (Electric)	Fixed (Constant)	Adjustable (Bio-stimulation / Hybrid)
SUSTAINABILITY	Low	Medium	High	Very High
INFRASTRUCTURE REQUIREMENT	High	High	Low	Low
LIGHT POLLUTION	High	Medium	Very Low	Very Low
OPERATIONAL COST	High	Medium	Very Low	Low
MAINTENANCE	High	Medium	Low	Low

Figure 3. Comparative analysis of the proposed invention and existing technologies.

D. DETAILED DESCRIPTION:

The Velocity-Responsive Bioluminescent Plant Matrix consists of three components that work together:

- (i) the light-emitting plant matrix,
- (ii) the velocity sensing component, and
- (iii) the control component linking both systems.

D1. Bioluminescent Plant Matrix – An arrangement of autoluminescent plant modules genetically modified for bioluminescence, grown in modular planter housing designed to be embedded into the ground at pathways, medians, or around buildings. Each module possesses an inherent bioluminescent pathway resulting in constant luminescence regardless of any stimuli. Modules are grouped in matrix segments and linked via a low-voltage biocompatible interface to actuation nodes.

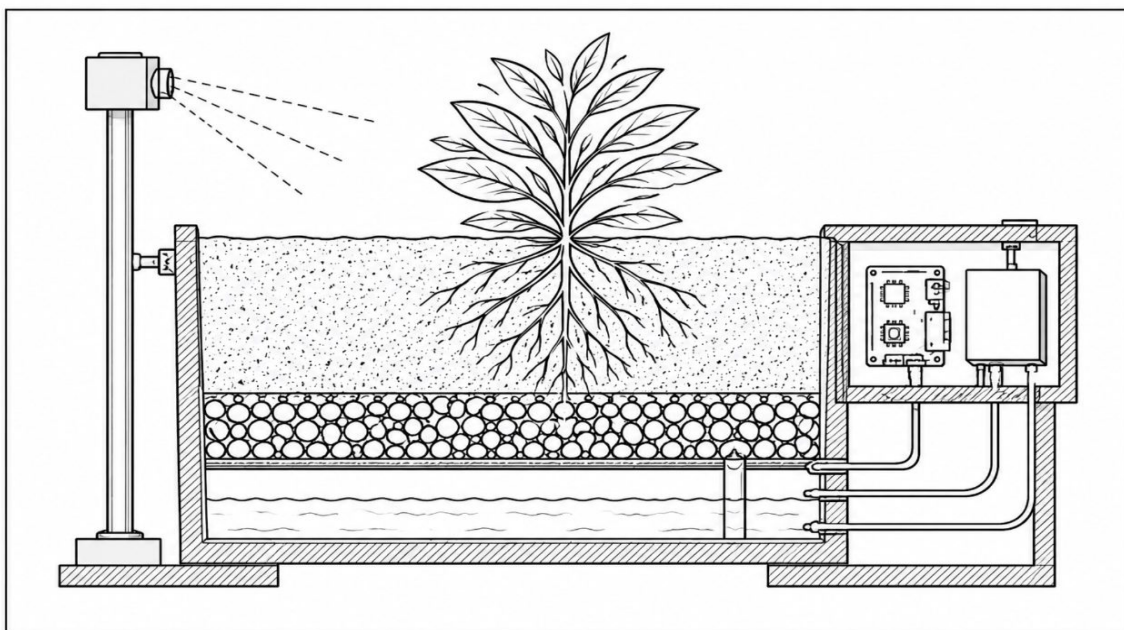


Figure 4. Cross-sectional view of a bioluminescent plant module.

D2. Velocity Detection Subsystem: Each segment is equipped with a non-invasive sensor (Doppler radar, passive-infrared array, or LIDAR-based ranging) that watches the approach zone and calculates approach velocity from successive range/angle readings when motion is detected.

D3. Modulation and Control Subsystem: A microcontroller within the segment cross-references the velocity calculation against a tiered response curve defining luminescence-boost parameters. Slow pedestrian approaches trigger localized increases in output from the relevant segment(s). Moderate cyclist approaches cause increases in a broad swath leading up to the anticipated arrival point, and fast vehicular approaches result in the maximum boost across the largest possible leading arc. The boost may be generated by either a) a biocompatible microinjection system delivering an appropriate luciferin-pathway substrate or b) a miniature auxiliary LED array, depending on the power constraints of the segment. The glow is then allowed to return to baseline over a programmed period after the last detected motion.

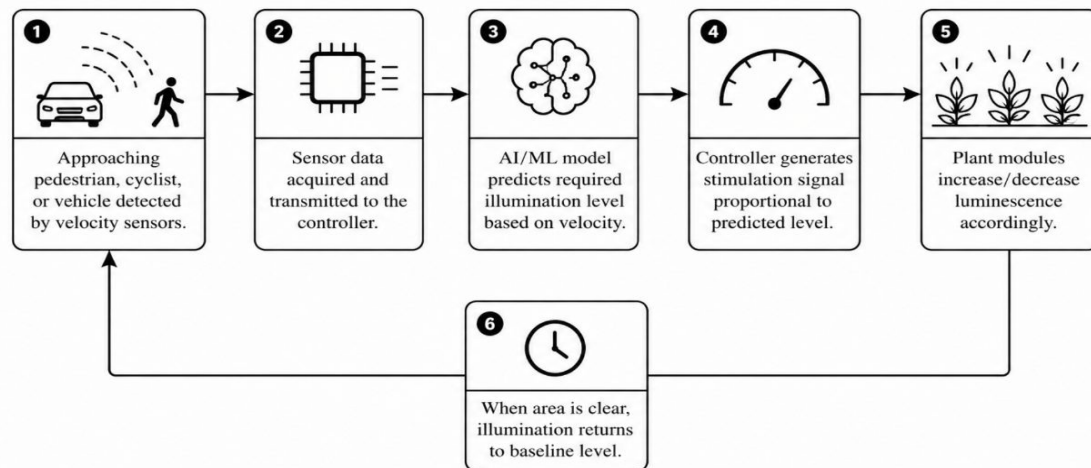


Figure 5. Operational workflow of the proposed system.

D4. Artificial Intelligence and Machine Learning Module: The proposed Velocity-Responsive Bioluminescent Plant Matrix comprises an AI and ML program that interprets the approach of pedestrians, cyclists, and motorists and responds by adjusting the lighting. Unlike conventional motion-activated lighting systems, the AI-based system predicts the movement patterns of the objects and determines the appropriate lighting level to provide the desired amount of illumination.

First, the AI subsystem receives movement information from the speed detection module in the form of distance, velocity, direction, detection timestamp, object size, and illumination level. The data acquisition process depends on the nature of the sensing hardware, which may comprise Doppler radar, LiDAR, passive IR sensors, ultrasonic sensors, or vision-based modules. The acquired data then undergoes preprocessing procedures like noise reduction and feature extraction before being fed to the ML algorithm. Denoising methods comprise the Kalman filter and moving average algorithms, while feature normalization prevents distortion caused by inconsistent input levels.

Next, the obtained features are used to predict the intensity, distribution, and time dynamics of the lighting bioluminescent matrix. The prediction model can also use the Random Forest Regression algorithm, which provides higher accuracy, less sensitivity to bad data, faster calculation, and simpler implementation, allowing it to be used on Raspberry Pi or ESP32 embedded systems. In addition, the model can be customized using different machine learning algorithms, such as XGBoost, Gradient Boosting, Artificial Neural Networks, or Support Vector Regression. For time-series pattern recognition, it is possible to use Long Short-Term Memory networks, which highlight the movement patterns as well as predict the required lighting conditions.

Finally, the model is trained with a dataset which captures the movement patterns of motorists, pedestrians, and cyclists operating at varying speeds in different environmental conditions. Using the features identified in the preprocessing step, a labeled training set is developed and used to develop the prediction model. Notably, the input ports of the algorithm comprise the speed, direction, distance, and object class, alongside the desired illumination level. After training, the model is stored in the embedded processor as a script

capable of executing prediction tasks on the local network without cloud-based computation.

Based on the predicted output, the AI modules categorized the incoming entity according to the predefined speed categories. Slow-foot traffic produced local low-intensity illumination, medium-speed cyclists prompted a larger area of moderate intensity, and a fast vehicle resulted in a broad bright spot to enhance the safety of the plant. The prediction outcome from the Machine Learning model is transferred to the illumination control sub-module that manages the biochemical stimulation circuit or the auxiliary low-power LED driver circuit that supports the visible bioluminescence intensity of the plants.

D5. Complete Working of the Proposed System:

The system constantly analyses the area surrounding the road with the velocity sensors. When any of the objects are identified, the information received by the sensors is sent to the embedded controller. Then, with the help of the AI model, it identifies the required lighting level based on the object's speed and direction of movement. After that, the controller activates the necessary plant modules or the additional lighting source. Thus, when the tracked object moves away, the level of illumination is progressively returned to the standard to save electricity.

D6. Prototype Implementation:

The prototype consists of genetically engineered bioluminescent plant modules integrated with Doppler radar or PIR sensors, an embedded controller such as Raspberry Pi or ESP32, communication interfaces, and a low-power stimulation mechanism. The AI model is deployed on the embedded controller to perform real-time prediction and adaptive illumination control. The prototype is intended for laboratory validation before large-scale commercial deployment.

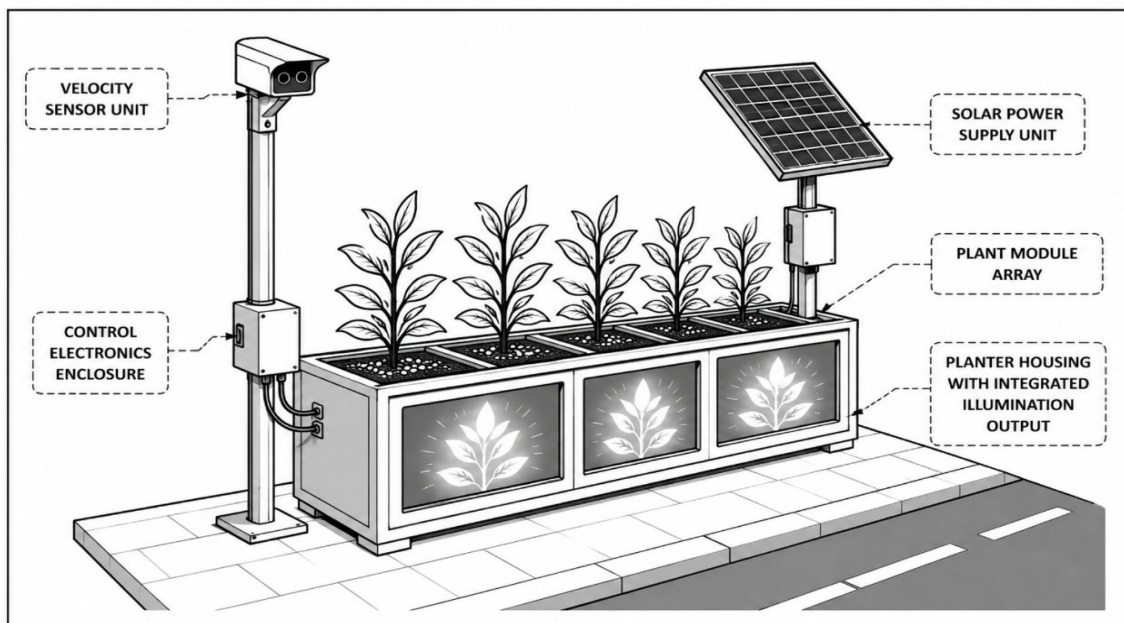


Figure 6. Prototype configuration of the proposed system.

D7. Commercial Applications:

The invention can be applied in intelligent cities, parks, universities, sidewalks, highways, airports, railway stations, hospitals, and housing estates. The lighting system can increase

security and save energy.

D8. Estimated Prototype Development Cost:

The estimated prototype development cost is about INR 6–8 lakhs, including genetic engineering of plant modules, sensing hardware, embedded controllers, software development, AI model training, prototype fabrication, and system testing. Commercial manufacturing is expected to reduce the overall production cost through large-scale deployment. (Bioluminescent Petunia to Light Gardens and Homes at Night, 2023)

E. RESULTS AND ADVANTAGES:

Expected Results:

A measurable increase in perceived segment brightness shortly after detection; a tiered, velocity-correlated difference in illumination breadth/intensity between pedestrian, cyclist, and vehicular triggers; reduced net auxiliary electrical draw versus an equivalent fully-electric luminaire array, owing to the constitutive baseline glow requiring no continuous electrical input. (These are stated as target outcomes for a system under prototype development, not measured results — revise with real lux, latency, and energy data once the prototype exists.)

Advantages:

1) Reduced grid-electricity dependency for low/moderate-intensity lighting; (2) a differentiated, velocity-proportional response mirroring adaptive electric lighting without requiring users to carry any device; (3) living green infrastructure combining urban greenery with functional lighting; (4) modular horticultural maintenance without rewiring; (5) potential reduction in light pollution.

F. EXPANSION:

Extension to indoor/building-perimeter bioluminescent façade elements responsive to foot traffic. Integration with municipal smart-city sensor networks for city-wide traffic/pedestrian heatmap data. Climate-specific plant variants for deployment through multiple urban geographies.

A fully passive (non-electrical) actuation pathway using only biochemical micro-dosing.

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:

The working prototype is under development.

H. EXISTING DATA: Any clinical or comparative data necessary to support your invention. (Comparative) NA

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. NA

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside the university (Any Industry, other Universities, or any other entity). NA

7. Potential Chances of Commercialization. Yes

8. List of companies which can be contacted for commercialization along with the website link. NA

9. Any basic patent which has been used and we need to pay royalty to them. NA

10. FILING OPTIONS: Please indicate the level of your work, which can be considered for provisional/complete/PCT filings. Complete

11. KEYWORDS: Glow-in-the-dark plants, autoluminescent phytosensor, synthetic biology, smart urban lighting, velocity-responsive illumination, adaptive street lighting, transgenic luciferase pathway, green infrastructure, energy-saving lighting, velocity sensing, modular plant matrix, eco-friendly urban design. Artificial Intelligence; Machine Learning; Edge Computing; Smart Infrastructure.

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Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
	US20050081268A1	Discloses a genetically-modified plant with a constitutive promoter driving a luciferase gene. No urban-lighting application.	Field plants have no urban-lighting application.	Field plants have no urban-lighting application.
II.	US7049483B1	The invention relates to transgenic plants containing a luciferase gene under the control of a lighting control architecture.	Transgenic plants contain a luciferase gene under no lighting control architecture.	Transgenic plants contain a luciferase gene under no lighting control architecture.
III.	US9284569	Discloses autoluminescent plants that can be used for various applications, including as a light source for night vision.	The plants have a light source for night vision.	The plants have a light source for night vision.
IV.	US9441236	The technology describes a method for creating a plant with a light-emitting protein that can be used for various applications, including as a light source for night vision.	One use of a light-emitting protein is as a light source for night vision.	One use of a light-emitting protein is as a light source for night vision.
V.	GB2455504A	The invention concerns a method for creating a plant with a light-emitting protein that can be used for various applications, including as a light source for night vision.	A method for creating a plant with a light-emitting protein that can be used for various applications, including as a light source for night vision.	A method for creating a plant with a light-emitting protein that can be used for various applications, including as a light source for night vision.
VI.	US6204615	Discloses an outdoor lighting system that uses a plant with a light-emitting protein as a light source.	The control system is a plant with a light-emitting protein as a light source.	The control system is a plant with a light-emitting protein as a light source.
VII.	US7864072	The technology describes a method for creating a plant with a light-emitting protein that can be used for various applications, including as a light source for night vision.	The technology describes a method for creating a plant with a light-emitting protein that can be used for various applications, including as a light source for night vision.	The technology describes a method for creating a plant with a light-emitting protein that can be used for various applications, including as a light source for night vision.

Have you described or shown your invention to anyone or in any conference?	YES	NO (✓)
Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing rights)?	YES	NO (✓)
Has your invention been described in any printed publication, or any other form of media (such as the Internet)?	YES	NO (✓)
Do you have any collaboration with any other institution or organization on the same?	YES	NO (✓)
Name of Regulatory body or any other applicable authority, if required.	NO	NO (✓)